The PCs and GSM phones are not allowed. Printed Notes/Books and the calculators can be used. ***All the answers must be clearly stated, otherwise no partial credit will be given. Please write your answers clearly visible: this is the responsibility of the student.*** *The duration is 110 minutes. When the exam is completed, please scan and upload your answer sheet to UES system:* ***USE PDF FORMAT.***

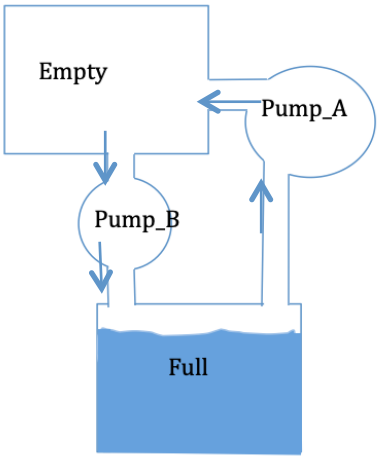
1. The state transition matrix for a discrete system is given as 'A'. Show that the system is not stable if the determinant value of the matrix 'A' is greater than one. (10 pts).
2. Find a non-zero function which is orthogonal to the following functions: 1, x^2, x^4. How many different functions satisfying the requirement can be found? Explain. (20 pts)

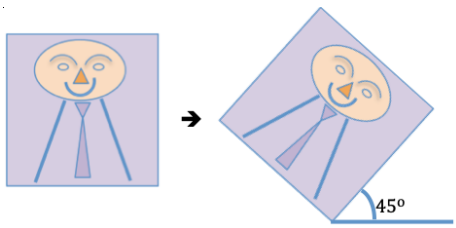
In question2 use the interval (0,1) for the functions.

3. The bitcoin/dolar change for the last months can be seen in the figure. The changes for march,april and may are given according to the bitcoin price at Feb 1st . Use a quadratic function to determine the change in June. (20 pts)



4. Consider the system giveb in the figure below, here tankA is full of water and tankB is empty. The flow directions for pumA and pumpB are given in the figure. PumpA can transfer whole water in tankA in 20 mins while pumpB transfers whole water inside tankB in 100 mins.
PumpA and pumpB are activated/deactivated in 10 minutes intervals:
(10 minutes on, 10 minutes off, 10 mins. on, 10 mins. off, ...).
In this problem, we want to analyze the water levels in tanka and tankB.
(50 pts.)
a. Find the differential equation set describing the system.
b. Write the state transition matrix.
c. Show the characteristic equation of the system
d. Determine the eigenvalues and eigen vectors
e. What will be the water levels in both tanks after 4 hours?
f. Find the steady state water levels.





5. Consider a linear tranformation that rotates an image by 45 degrees. Determine the transformation matrix.
(If the optimum basis that results in the simplest form of transformation matrix is used you will have 20 points otherwise 5 pts.)